RIGEL: Self-Distilled Score Adaptation for Image and Video Captioning Evaluation

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Abstract

Automatic evaluation of image and video captioning is essential for benchmarking multimodal systems, although standard evaluation metrics show limited alignment with human judgments. Recent approaches using large language models (LLMs), commonly referred to as LLM-as-a-Judge, have improved alignment with human judgments but still suffer from a mismatch between large-vocabulary language modeling and evaluation over a small label set. To address this, we propose RIGEL, an automatic evaluation metric for image and video captioning, based on self-distilled score adaptation. The metric employs an evaluation-specific scoring head distilled from a frozen LLM, which captures judgment signals in a task-aligned space without relying on large-vocabulary token sets. We then refine the LLM backbone with human judgment data. To train RIGEL, we constructed the Vid-Lepus dataset, which contains 3,338 video clips, 33,380 reference captions, and 5,637 candidate captions. Experiments on multiple benchmarks show that RIGEL outperforms state-of-the-art metrics, achieving over 10-point improvements on ActivityNet-Fact in the reference-free setting. Our project page is available at https://rigel-mnghv.kinsta.page/

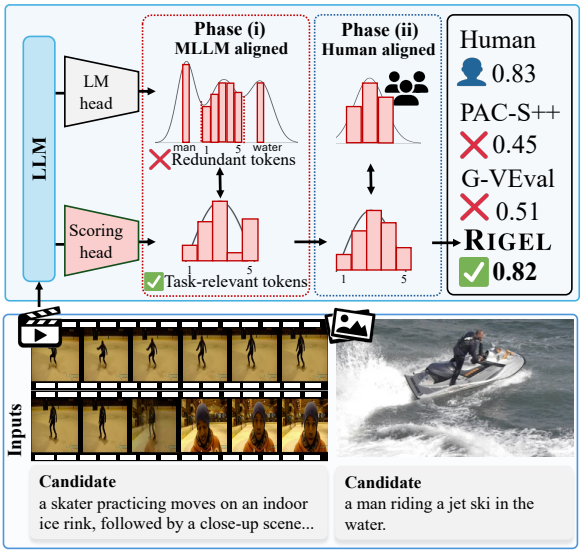


Figure 1: **Overview of RIGEL.** A two-phase framework for human-aligned caption evaluation. In Phase 1, an evaluation-specific scoring head is distilled from a frozen large language model (LLM) to map hidden representations to ordinal judgment scores, alleviating the mismatch between the LM vocabulary and the ordinal label set in the original language modeling (LM) head. In Phase 2, the LLM backbone is refined using human judgments (e.g., scores ranging from 1 to 5) while freezing the scoring head's parameters, yielding task-aligned evaluations.

1 Introduction

Automatic evaluation of image and video captioning is essential for benchmarking multimodal systems (Bai et al., 2023; Chen et al., 2024; Team et al., 2023). However, standard evaluation metrics show limited alignment with human judgments. Traditional metrics such as BLEU (Papineni et al., 2002) and METEOR (Banerjee and Lavie, 2005) rely on lexical or semantic matching and correlate weakly with human judgments (Hessel et al., 2021; Sarto et al., 2023, 2025). Data-driven metrics based on vision language models (e.g., Polos (Wada et al., 2024)) improve correlation but capture surface-level similarity rather than fine-grained dimensions such as descriptiveness, relevance, and fluency.

Recent LLM-based approaches such as FLEUR (Lee et al., 2024) and G-VEval (Tong et al., 2025) yield more interpretable judgments (Gu et al., 2025). However, these methods are limited by their reliance on predefined token sets for scoring. The language modeling (LM) head operates over a large vocabulary set $\mathcal{V}$ ($|\mathcal{V}| \sim 10^5$), while evaluation requires prediction over a small ordinal label set $\mathcal{M}$ ($|\mathcal{M}| \ll |\mathcal{V}|$). The distributions of LM-head logits assigned to score tokens ("1"–"5") and the remaining vocabulary are presented in Fig. 2. We observe that the logits

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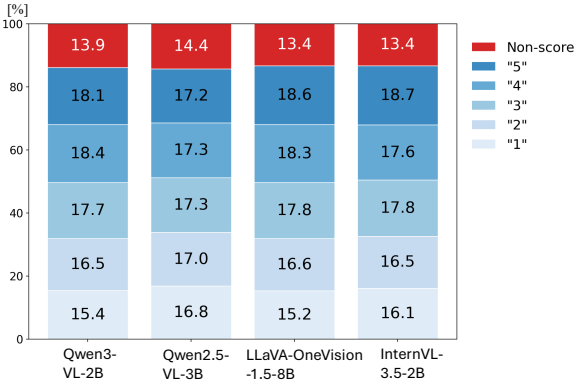


Figure 2: Logit distributions over score tokens ("1"–"5") and non-score tokens on the Spica dataset (Hirano et al., 2026). Non-score tokens exhibit logit magnitudes comparable to those of score tokens across the four models Qwen3-VL-2B (Bai et al., 2025a), Qwen2.5-VL-3B (Bai et al., 2025b), LLaVA-OneVision-1.5-8B (An et al., 2025), and InternVL-3.5-2B (Wang et al., 2025). This observation provides partial evidence for our claim that non-score tokens function as noise in score prediction. The implementation details are provided in Appendix B.

assigned to task-irrelevant tokens are comparable in magnitude to those of the score tokens. This indicates that the LM head predicts scores while assigning many logits to task-irrelevant tokens, which can introduce noise and degrade evaluation performance.

Based on the above analysis, we propose RIGEL, an automatic evaluation metric for image and video captioning, which addresses the misalignment between the LM vocabulary and the ordinal label set through self-distilled score adaptation framework (Fig. 1). The framework consists of the following two phases: First, the metric employs an evaluation-specific scoring head distilled from a frozen LLM, which captures judgment signals in a task-aligned space without relying on large vocabulary token sets. Second, we refine the LLM backbone with human judgment data while freezing the head's parameters. Our empirical results show that each phase contributes independently to the final performance (Section 5.3).

To train our proposed metric, training data that pair candidate captions with human judgments are required. However, existing datasets (Liu and Wan, 2023; Shi et al., 2022) often lack this form of paired supervision, making it challenging to develop supervised evaluation metrics. To address this limitation, we constructed Vid-Lepus, a dataset for training video captioning evaluation metrics. Vid-Lepus comprises video clips, reference captions, candidate captions, and corresponding human judgments.

The main contributions of this study are summarized as follows:

- We propose RIGEL, a unified evaluation metric for image and video captioning (Section 3), addressing the vocabulary-label mismatch between the LM head vocabulary and ordinal label set.
- We introduce self-distilled score adaptation framework. Our empirical results show that our metric, RIGEL, outperforms existing metrics on standard benchmarks (Section 5).
- We introduce Vid-Lepus, a dataset for training human-aligned metrics for video captioning, featuring $3,338$ video clips, $33,380$ reference captions, $5,637$ candidate captions, and $14,802$ human judgments (Section 4).

2 Related Work

Image captioning metrics. Automatic evaluation metrics for image captioning, such as BLEU (Papineni et al., 2002), METEOR (Banerjee and Lavie, 2005), ROUGE (Lin, 2004), CIDEr (Vedantam et al., 2015), and SPICE (Anderson et al., 2016), along with extensions such as CIDEr-R (Oliveira et al., 2021) and JaSPICE (Wada et al., 2023), have traditionally relied on reference-based lexical or semantic matching. However, these metrics often correlate only weakly with human judgments, particularly when captions are semantically correct but lexically diverse (Hessel et al., 2021; Sarto et al., 2023, 2025).

To address this limitation, recent work has proposed data-driven metrics that use pre-trained vision–language models or multimodal encoders (Lee et al., 2020, 2021; Sarto et al., 2023). Among them, Polos (Wada et al., 2024) incorporates image information and supervised learning from human judgments. These metrics have been shown to align well with human judgments on standard image captioning evaluation benchmarks, most of which primarily contain short captions.

LLM-as-a-Judge approaches. Recent advances in LLMs and multimodal large language models (MLLMs) have led to a new family of evaluation metrics, often referred to as LLM-as-a-Judge approaches (Gu et al., 2025). These metrics evaluate captions using LLMs or MLLMs, often yielding more interpretable judgments than embedding-based similarity metrics. Several such

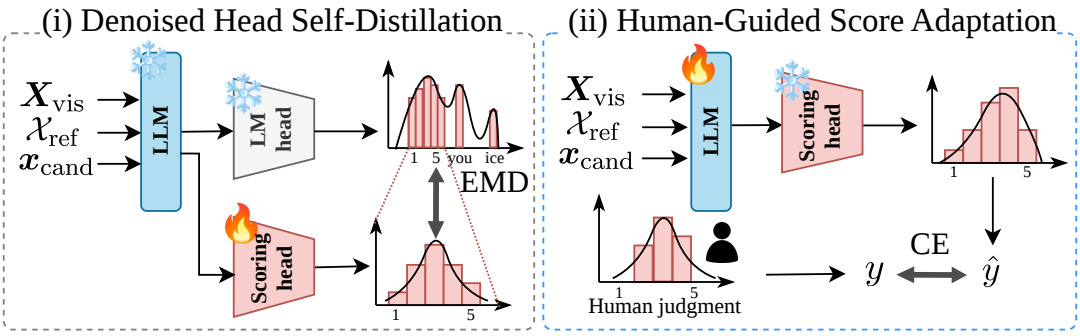


Figure 3: **Overview of our proposed two-phase training framework.** (i) Scoring head (red block) is trained with five labels using Earth Mover's Distance (EMD) while the LLM and the LM head are frozen. (ii) The LLM backbone is fine-tuned using human judgments while freezing the scoring head's parameters. CE represents cross-entropy.

approaches have been proposed for image captioning. For example, FLEUR (Lee et al., 2024) employs LLaVA (Liu et al., 2023, 2024) to incorporate image inputs directly for caption evaluation. Similarly, G-VEval (Tong et al., 2025) and HarmonicEval (Ohi et al., 2024) score captions from multiple perspectives, improving interpretability and alignment with human evaluation criteria.

Despite their promising performance, LLM-based metrics suffer from a practical drawback: their scores are computed from LM-head logits over the full vocabulary, while caption evaluation only requires a small set of ordinal labels. As a result, much of the logit mass is assigned to task-irrelevant tokens.

Video captioning metrics. Compared with image captioning, automatic evaluation for video captioning has received less attention. EMScore (Shi et al., 2022) extends embedding-based matching to the video domain; PAC-S and PAC-S++ (Sarto et al., 2023, 2025) compute CLIP-based similarity adapted for captioning evaluation; FactVC (Liu and Wan, 2023) evaluates factual consistency between video and candidate captions; and G-VEval (Tong et al., 2025) applies the LLM-as-a-Judge paradigm to video captioning.

Datasets and benchmarks. A variety of datasets have been used to evaluate image captioning metrics, including Composite (Aditya et al., 2015), Flickr8K-Expert and Flickr8K-CF (Hodosh et al., 2013), Polaris (Wada et al., 2024), and Nebula (Matsuda et al., 2024). These benchmarks have played an important role in analyzing metric quality, but many of them provide human judgments from a single evaluation perspective, limiting their ability to assess fine-grained aspects of caption quality. This limitation is important for short captions, where minor wording differences can affect multiple quality dimensions simultaneously.

To provide fine-grained supervision, recent work has introduced datasets with multi-dimensional human judgments (Ohi et al., 2024; Kasai et al., 2022). For video captioning, the VATEX-EVAL and ActivityNet-FOIL benchmarks (Shi et al., 2022), as well as the ActivityNet-Fact and YouCook2-Fact benchmarks (Liu and Wan, 2023), have been used to evaluate metrics. However, they lack training sets with human judgments, making it difficult to develop supervised evaluation metrics. To address this limitation, we constructed Vid-Lepus (Section 4), a dataset with human judgments for training supervised video captioning evaluation metrics.

3 Method

We propose RIGEL, an automatic evaluation metric for both image and video captioning. This metric improves LLM-based evaluation through self-distilled score adaptation framework, and it resolves the mismatch between large-vocabulary language modeling and evaluation tasks. This scheme enables efficient evaluation with a single forward pass. Fig. 3 shows an overview of the proposed framework, which consists of two phases: (i) Denoised Head Self-Distillation and (ii) Human-Guided Score Adaptation. The proposed self-distilled score adaptation can be broadly applied to existing LLM-as-a-Judge frameworks. Section 3.1 describes Phase 1, which learns an evaluation-specific scoring head via self-distillation, Section 3.2 describes Phase 2, which adapts the LLM backbone to human judgments, and Section 3.3 presents the inference procedure.

We define the input to the metric as a triplet $(\boldsymbol{X}_{\text{vis}}, \mathcal{X}_{\text{ref}}, \boldsymbol{x}_{\text{cand}})$, where $\boldsymbol{X}_{\text{vis}} \in \mathbb{R}^{T\times 3\times H\times W}$ is

a visual input consisting of T frames of RGB images with height H and width W (with $T = 1$ for still images), $\mathcal{X}_{\text{ref}} = \{\boldsymbol{x}_{\text{ref}}^{(i)}\}_{i=1}^{N}$ is a set of N reference captions, and $\boldsymbol{x}_{\text{cand}}$ is a candidate caption. In the reference-based setting, $\mathcal{X}_{\text{ref}}$ consists of N reference captions, whereas in the reference-free setting, $\mathcal{X}_{\text{ref}} = \emptyset$. Given these inputs, the proposed metric outputs the final evaluation score $\hat{y}$.

3.1 Phase 1: Denoised Head Self-Distillation

Phase 1 introduces an evaluation-specific scoring head distilled from a frozen LLM. This head captures the model's judgment signals in a task-aligned label space via self-distillation. We assume that the original LM head, which has been optimized for next-token prediction over a large vocabulary, is not well suited for predicting a small, predefined set of ordinal labels. This mismatch can make the LM head unsuitable for direct use in evaluation. To bridge this gap, we train a lightweight scoring head through self-distillation, transferring the LLM's evaluation capability from the noisy vocabulary space into a clean ordinal prediction space.

Given an input, we construct a prompt and process it with a frozen LLM to obtain a final-layer hidden representation $\boldsymbol{h} \in \mathbb{R}^d$, where d denotes the hidden dimension. From the vocabulary logits, we extract those corresponding to the ordinal score tokens and denote them by $\boldsymbol{z}_{\text{score}} \in \mathbb{R}^{|\mathcal{M}|}$, where $\mathcal{M}$ denotes an ordinal label set; following previous studies (Wada et al., 2024; Matsuda et al., 2024; Shi et al., 2022), we set $\mathcal{M} = \{1, 2, 3, 4, 5\}$. We then apply a temperature-scaled softmax to obtain a soft pseudo-label distribution:

$$\boldsymbol{q} = \text{softmax}(\boldsymbol{z}_{\text{score}}/\tau), \tag{1}$$

where τ is the distillation temperature.

We replace the LM head with a lightweight MLP scoring head $g_\theta : \mathbb{R}^d \to \mathbb{R}^{|\mathcal{M}|}$, which produces denoised logits: $\hat{\boldsymbol{z}} = g_\theta(\boldsymbol{h})$. We train g_θ by minimizing the Phase 1 distillation loss $\mathcal{L}_{\text{P1}}$, defined as follows:

$$\mathcal{L}_{\text{P1}} = \tau \, \text{EMD}(\boldsymbol{q}, \text{softmax}(\hat{\boldsymbol{z}}/\tau)), \tag{2}$$

where EMD denotes the one-dimensional Earth Mover's Distance (EMD) over the ordered score labels (Rubner et al., 2000), defined as follows:

$$\text{EMD}(\boldsymbol{a}, \boldsymbol{b}) = \sum_{i=1}^{K-1} \left| \sum_{j=1}^{i} a_j - \sum_{j=1}^{i} b_j \right|, \tag{3}$$

where $\boldsymbol{a}$ and $\boldsymbol{b}$ denote probability distributions over K ordered labels, and a_j and b_j denote the probabilities assigned to the j-th score label. The factor τ is used to keep the gradient scale stable when applying temperature scaling. During this phase, the LLM remains frozen and only the scoring head parameters θ are optimized.

3.2 Phase 2: Human-Guided Score Adaptation

Phase 2 adapts the LLM backbone to human judgments under the frozen scoring head g_{θ^*} learned in Phase 1, where θ^* denotes the scoring head parameters obtained in Phase 1. We keep g_{θ^*} frozen and fine-tune only the backbone parameters via Low-Rank Adaptation (LoRA) (Hu et al., 2022), stabilizing training. The backbone learns representations aligned with the scoring head.

Given a one-hot gold label $\boldsymbol{y}^{\text{gold}} \in \{0, 1\}^{|\mathcal{M}|}$ derived from averaged human judgment, we minimize the Phase 2 cross-entropy loss $\mathcal{L}_{\text{P2}}$ as follows:

$$\mathcal{L}_{\text{P2}} = -\sum_{k=1}^{|\mathcal{M}|} y_k^{\text{gold}} \log p_k, \tag{4}$$

where p_k denotes the k-th element of $\text{softmax}(g_{\theta^*}(\boldsymbol{h}))$. Only the LoRA parameters of the backbone are updated and θ^* remains fixed.

3.3 Inference

At inference, we compute $\hat{y}$ via score smoothing (Lee et al., 2024; Tong et al., 2025): $\hat{y} = \sum_{k=1}^{|\mathcal{M}|} k \cdot p_k$, where $p_k = \text{softmax}(g_{\theta^*}(\boldsymbol{h}))_k$. We linearly normalize $\hat{y}$ to the range [0,1].

4 Vid-Lepus Dataset

Supervised metrics for image and video captioning require a training dataset that includes candidates annotated with human judgments. Existing video-captioning evaluation datasets are insufficient because they do not provide training data with human judgments (Liu and Wan, 2023; Shi et al., 2022). To address this limitation, we constructed Vid-Lepus, a supervised video captioning dataset, for training evaluation metrics. This dataset comprises video clips, references, and candidate captions paired with human judgments.

From crowd workers, we collected human judgments on a five-point scale to evaluate the appropriateness of candidate captions with respect to the video clips and references. Annotators were

Methods	Flickr8K-Expert		Flickr8K-CF		Nebula		Composite		FOIL	
	τ_b	τ_c	τ_b	τ_c	τ_b	τ_c	τ_b	τ_c	1-ref[%]	4-ref[%]
Reference-based										
BLEU	30.6	30.8	16.4	8.7	46.5	44.1	28.3	30.6	66.5	82.6
ROUGE	32.1	32.3	19.9	10.3	45.8	43.4	30.0	32.4	71.7	79.3
CIDEr	43.6	43.9	24.6	12.7	51.5	48.8	34.9	37.7	82.5	90.6
METEOR	41.5	41.8	22.2	11.5	50.2	47.6	36.0	38.9	78.8	82.6
SPICE	51.7	44.9	24.4	12.0	51.5	47.4	38.8	40.3	75.5	86.1
BERTScore	37.8	46.7	22.8	11.5	47.5	47.1	30.2	30.1	88.6	92.1
RefCLIP-S	52.6	53.0	36.4	18.8	53.6	50.8	51.2	55.4	91.0	92.6
RefPAC-S	55.5	55.9	37.6	19.5	54.7	51.9	53.0	57.3	93.7	94.9
Polos	56.1	56.4	37.8	19.5	58.0	55.0	53.7	57.6	93.3	95.4
Ref-HICEScore	57.2	57.7	38.2	19.8	-	-	53.9	58.7	96.4	97.0
DENEB	55.6	56.5	38.0	19.6	58.1	55.1	54.0	57.9	95.1	96.1
RefPAC-S++	55.3	55.7	37.9	19.6	53.3	50.6	54.7	59.1	93.5	94.1
Pearl	58.2	58.6	38.6	20.0	58.4	55.4	55.8	60.4	96.5	97.2
HiFiScore	-	58.4	-	-	-	-	-	65.8	-	-
CLAIR	58.3	48.8	38.2	17.0	-	-	-	61.0	-	93.6
Ref-FLEUR	-	51.9	38.8	-	-	-	-	64.2	97.3	98.4
G-VEval	**60.5**	58.7	38.2	19.9	-	-	-	-	97.8	98.4
Ours	58.6	**59.0**	**40.4**	**20.9**	**60.6**	**57.5**	**61.2**	**66.1**	**99.1**	**99.2**
Reference-free										
CLIP-S	51.1	51.2	34.4	17.7	50.5	47.9	49.8	53.8	87.2	87.2
PAC-S	53.9	54.3	36.0	18.6	51.0	48.3	51.5	55.7	89.9	89.9
BRIDGE	55.4	55.8	36.3	19.0	-	-	52.9	57.2	93.0	93.0
HICEScore	55.9	56.4	37.2	19.2	-	-	53.1	57.9	93.1	93.1
PAC-S++	54.1	54.5	37.0	19.1	50.5	47.9	53.9	58.3	90.2	90.2
BLIP2-Score	52.2	52.5	36.7	19.0	53.0	50.7	56.9	61.5	94.3	94.3
Pearl	56.2	56.6	37.8	19.5	55.9	53.0	54.0	58.4	96.7	96.7
HiFiScore	-	58.4	-	-	-	-	-	65.7	-	-
FLEUR	-	53.0	38.6	-	-	-	-	63.5	96.8	96.8
G-VEval	**61.5**	59.7	38.7	20.2	-	-	-	-	-	-
EXPERT	-	56.7	39.3	-	-	54.9	-	65.0	-	-
DISCODE (LLaVA)	55.7	56.1	40.2	20.8	-	-	**61.1**	**66.0**	-	-
DISCODE (InternVL)	57.7	58.1	40.1	20.8	-	-	60.5	64.9	98.2	98.2
Ours	59.5	**59.9**	**40.4**	**20.9**	**58.1**	**55.1**	59.5	64.3	**98.7**	**98.7**

Table 1: **Quantitative comparison on image captioning evaluation benchmarks. Bold** font indicates the best results, and underlining indicates the second-best results. Following previous work (Hirano et al., 2026; Tong et al., 2025; Lee et al., 2024), we report the reproduced results for CLAIR. "-" indicates either non-executable code or unavailable data.

instructed to assess the candidates across three dimensions: descriptiveness, relevance, and fluency, following prior work (Wada et al., 2024; Matsuda et al., 2025). For quality control, we excluded annotations from evaluators exhibiting suspicious behavior, such as unusually short response times or consistently uniform ratings. Furthermore, samples whose human ratings exhibited a range of at least 3 on the five-point scale are manually reviewed by an expert annotator to resolve annotation disagreements.

To assess annotation reliability, we computed Krippendorff's α for inter-annotator agreement, which was 0.45. This value was higher than those of standard datasets, including CapEval1K (0.37) (Lee et al., 2021) and Spica (0.44) (Hirano et al., 2026). Further details on the dataset and its construction process are provided in Appendix D.

5 Experiments

5.1 Experimental Setup

Datasets. For the image captioning evaluation, we used the standard benchmarks Flickr8K-Expert (Hodosh et al., 2013), Flickr8K-CF (Hodosh et al., 2013), Nebula (Matsuda et al., 2024), Composite (Aditya et al., 2015), and FOIL (Shekhar et al., 2017). To evaluate the video captioning metric, we used VATEX-EVAL (Shi et al., 2022), ActivityNet-FOIL (Shi et al., 2022), ActivityNet-Fact (Liu and Wan, 2023), and YouCook2-Fact (Liu and Wan, 2023). To train RIGEL, we used the constructed Vid-Lepus dataset for video captioning along with Spica (Hirano et al., 2026) for image captioning. The Spica dataset (Hirano et al., 2026) was used with its original training and validation splits, containing

Metrics	VATEX-EVAL				ActivityNet-Fact			YouCook2-Fact			ActivityNet-FOIL
	1-ref		9-ref		Para	Sent	Word	Para	Sent	Word	Accuracy [%]
	τ_b	ρ	τ_b	ρ	r	r	r	r	r	r	
Reference-based											
EMScore	28.6	37.1	36.8	47.2	42.7	35.2	44.6	54.3	51.4	55.3	92.4
RefPAC-S	31.4	40.5	38.1	48.8	47.0	37.8	49.5	56.2	54.4	57.0	93.5
RefPAC-S++	32.2	41.5	39.8	50.8	-	-	-	-	-	-	93.4
FactVC	-	-	-	-	55.1	46.5	54.5	60.6	58.3	61.5	-
G-VEval	44.9	-	48.1	-	56.6	48.1	48.8	**72.4**	**66.7**	**62.4**	-
Ours	**45.6**	**57.8**	**50.8**	**64.0**	**67.8**	**54.0**	**65.7**	65.4	60.8	61.7	**97.1**
Reference-free											
EMScore	23.2	30.3	23.2	30.3	25.3	19.0	30.0	33.7	35.3	36.1	89.5
PAC-S	25.1	32.6	25.1	32.6	33.2	27.1	38.4	31.2	33.5	33.1	90.1
PAC-S++	28.1	36.4	28.1	36.4	-	-	-	-	-	-	91.0
FactVC	-	-	-	-	46.2	37.1	48.0	40.8	41.0	42.0	-
G-VEval	39.4	-	39.4	-	48.8	38.6	43.6	**55.8**	50.1	44.4	-
Ours	**42.1**	**53.7**	**42.1**	**53.7**	**63.6**	**51.9**	**60.5**	55.0	**50.3**	**50.0**	**96.2**

Table 2: **Quantitative comparison on video captioning evaluation benchmarks.** "ρ" and "r" represent Spearman's and Pearson's correlation coefficients, respectively. Bold indicates the best result and underlining indicates the second-best result in each column.

Metrics	Phase 1	Phase 2	Head	Backbone	Composite		Flickr8K-EX		Nebula		VATEX-Eval	
					τ_b	τ_c	τ_b	τ_c	τ_b	τ_c	τ_b	ρ
(i)		✓	LM head	Qwen3-VL-2B	58.3	63.0	59.0	59.4	57.4	54.4	39.7	50.8
(ii)	✓		Scoring head	Qwen3-VL-2B	53.1	57.4	22.3	22.6	52.2	55.1	**43.2**	**54.3**
(iii)		✓	Scoring head	Qwen3-VL-2B	58.0	62.8	58.8	59.2	56.3	53.4	35.4	45.4
(iv)	✓	✓	Scoring head	Qwen2.5-VL-3B	59.4	64.2	59.2	59.7	57.3	54.3	39.8	50.8
(v)	✓	✓	Scoring head	Qwen3-VL-2B	**59.5**	**64.3**	**59.5**	**59.9**	**58.1**	**55.1**	42.1	53.7

Table 3: **Comparison across different training phases, heads, and backbones in the reference-free setting. Bold** indicates the best value in each column, and underlining indicates the second-best value. These results demonstrate that both Phases 1 and 2 contributed to the performance improvement on most benchmarks.

296, 149 and 3, 000 samples, respectively. The VidLepus dataset was divided into training and validation sets containing 13, 262 and 1, 540 samples, respectively. The training and validation sets of the datasets were used for metric training and hyperparameter tuning, respectively. Details of the datasets are provided in Appendix D.

Baselines. We selected multiple standard metrics for captioning evaluation as our baseline comparison metrics. For both image and video captioning, we used embedding-based metrics PAC-S (Sarto et al., 2023) and PAC-S++ (Sarto et al., 2025); and the LLM-as-a-Judge metric G-VEval (Tong et al., 2025). For image captioning, we additionally included traditional reference-based lexical matching metrics, namely BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), METEOR (Banerjee and Lavie, 2005), CIDEr (Vedantam et al., 2015), and SPICE (Anderson et al., 2016). We also included representative embedding-based metrics, including CLIPScore (Hessel et al., 2021), BERTScore (Zhang et al., 2020), HICEScore (Zeng et al., 2024a), BRIDGE (Sarto et al., 2024), and BLIP2-Score (Zeng et al., 2024b); supervised metrics trained on human judgments, including Polos (Wada et al., 2024), DENEB (Matsuda et al., 2024), and Pearl (Hirano et al., 2026); and LLM-as-a-Judge metrics, including CLAIR (Chan et al., 2023), FLEUR (Lee et al., 2024), HiFiScore (Yao et al., 2024), EXPERT (Kim et al., 2025), and DISCODE (Inoue et al., 2026). For video captioning, we further adopted two representative metrics in this field: EMScore (Shi et al., 2022) and FactVC (Liu and Wan, 2023). All experiments were reported based on a single run.

Evaluation metrics. We followed the standard evaluation practice used for each benchmark. Specifically, we used Kendall's τ_b and τ_c for the Composite, Flickr8K-Expert, Flickr8K-CF, and Nebula datasets; accuracy for FOIL and ActivityNet-FOIL; Pearson's correlation coefficient r for ActivityNet-Fact and YouCook2-Fact;

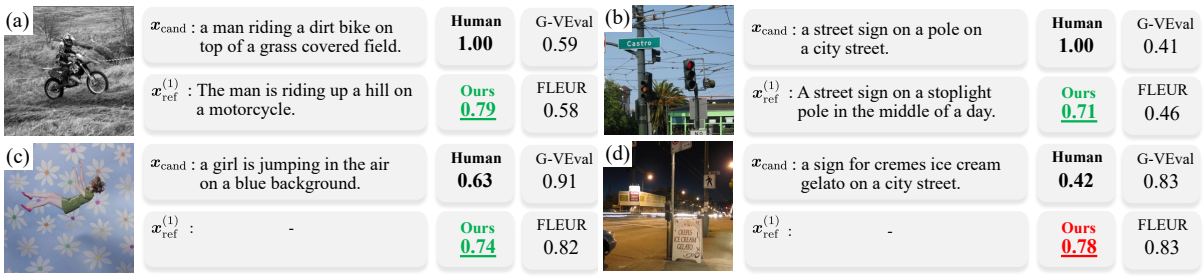


Figure 4: **Qualitative results on the Nebula dataset.** Cases (a)–(b) illustrate successful examples in the reference-based setting, whereas (c) shows a successful example in the reference-free setting. In contrast, (d) represents a failure case in the reference-free setting. Green values indicate predictions closest to human annotations, and red values denote critical errors. "-" indicates that no reference caption was provided.

and Kendall's τ_b together with Spearman's ρ for VATEX-EVAL.

5.2 Quantitative Results

Image captioning evaluation. Table 1 presents a quantitative comparison with baseline metrics on the Composite, Flickr8K-Expert, Flickr8K-CF, Nebula, and FOIL datasets. In the reference-based setting, our metric achieved strong performance across most benchmarks. On the Composite dataset, our metric outperformed the previous best baseline, and on FOIL, our metric achieved the highest accuracy in both the 1-ref and 4-ref settings. In the reference-free setting, our metric yielded competitive results, outperforming the baselines on several benchmarks including Flickr8K-CF, Nebula, and FOIL. These results demonstrate that our self-distilled score adaptation effectively improves the correlation with human judgments across image captioning benchmarks.

Video captioning evaluation. Table 2 presents a quantitative comparison of metrics on VATEX-EVAL, ActivityNet-Fact, YouCook2-Fact, and ActivityNet-FOIL. On VATEX-EVAL, our metric in the reference-based setting achieved improvements over the baselines in both τ_b and ρ across all reference settings (1-ref, 9-ref). On ActivityNet-FOIL, our metric achieved the highest accuracy, outperforming the best baseline score by a substantial margin. Table 2 shows that our proposed metric achieved the best Pearson's r on ActivityNet-Fact at the paragraph, sentence, and word levels. Similarly, in the reference-free setting, our metric outperformed all baseline metrics across most benchmarks, confirming the effectiveness of our approach for video captioning evaluation.

5.3 Ablation Study

Table 3 shows the quantitative results of the ablation studies. We conducted three ablation studies to investigate the contribution of each component in our proposed metric: Phase 1, Phase 2, and the backbone model.

Denoised Head Self-Distillation Ablation. We investigated the contribution of Phase 1 by excluding it from the full pipeline. As shown in Table 3, a comparison between Metric (iii), which has a randomly initialized scoring head, and Metric (v) indicates that excluding Phase 1 decreased the scores on all benchmarks. Specifically, Metric (v) outperformed Metric (iii) by 1.5 points on Composite, 0.7 points on Flickr8K-EX, and 1.8 and 1.7 points on Nebula for τ_b and τ_c, respectively. On VATEX-Eval, Metric (v) outperformed Metric (iii) by 6.7 and 8.3 points for τ_b and ρ, respectively. These results demonstrate that Phase 1 consistently improves performance. Furthermore, we compared Metrics (i) and (v) to assess the effect of using a scoring head within the full pipeline instead of the LM head. The results indicate that the scoring head is more effective than the LM head across all benchmarks. Specifically, Metric (v) outperformed Metric (i) by 1.2 and 1.3 points on Composite, 0.5 and 0.5 points on Flickr8K-EX, 0.7 and 0.7 points on Nebula for τ_b and τ_c, respectively.

Human-Guided Score Adaptation Ablation. We investigated the contribution of Phase 2 by excluding it from the full pipeline. As shown in Table 3, a comparison between Metrics (ii) and (v) indicates that excluding Phase 2 degrades the performance. Specifically, Metric (v) outperformed Metric (ii) by 6.4 and 6.9 points on Composite, 37.2 and 37.3 points on Flickr8K-EX for τ_b and τ_c, respectively. These results demonstrate that Phase



Figure 5: **Examples of successful cases from the VATEX-EVAL dataset.** Case (a) illustrates a successful example in the reference-based setting, whereas (b) shows a successful example in the reference-free setting.

2 substantially improves performance.

Backbone Ablation. We investigated the effect of the backbone model by replacing Qwen3-VL-2B (Bai et al., 2025a) with Qwen2.5-VL-3B (Bai et al., 2025b). As shown in Table 3, Metric (v) outperformed Metric (iv) by 0.3 and 0.2 points on Flickr8K-Expert, 0.8 and 0.8 points on Nebula for τ_b and τ_c, and 2.3 and 2.9 points on VATEX-Eval for τ_b and ρ. These results indicate that Qwen3-VL-2B performs at least as well as Qwen2.5-VL-3B and improves the results on Flickr8K-Expert, Nebula, and VATEX-Eval.

5.4 Qualitative Results

Image captioning. Fig. 4 presents representative examples of the results of the proposed metric on the Nebula dataset. We used Nebula for the qualitative analysis because it is a diverse and balanced dataset (Matsuda et al., 2024). Cases (a) and (b) show successful cases in the reference-based setting, while Case (c) displays a successful case in the reference-free setting. Case (d) illustrates a sample for which the proposed metric did not perform as expected.

In Case (a), $\boldsymbol{x}_{\text{ref}}^{(1)}$ was "the man is riding up a hill on a motorcycle." while x_{cand} was "a man riding a dirt bike on top of a grass covered field." In this sample, the human judgment was 1.00 as x_{cand} appropriately describes the image. However, G-VEval and FLEUR evaluated it as 0.59 and 0.58, respectively, while the proposed metric rated it as 0.79. Similarly, in Case (b), the metric yielded the score most aligned with human judgment.

In Case (c), $\boldsymbol{x}_{\text{cand}}$ was "a girl is jumping in the air on a blue background." For this caption, the average human judgment score was moderate at 0.63, because the caption mischaracterizes the girl as jumping. While G-VEval, FLEUR, and the original Qwen3-VL-2B using its LM head (Original LLM) provided scores of 0.91, 0.82, and 0.41, respectively, the proposed metric evaluated it at 0.74, closely aligned with the human judgment.

In Case (d), $\boldsymbol{x}_{\text{cand}}$ contains critical errors: it misread the text on the sign, rendering "crepes" as "cremes" and also failed to mention the background. Therefore, the average human judgment score was moderate at 0.42. However, our proposed metric gave a relatively high score of 0.78, indicating a discrepancy with respect to human judgment. Similarly, G-VEval, FLEUR, and the original LLM output scores of 0.83, 0.83, and 0.72, respectively. These results indicate that these metrics may not adequately penalize textual errors in scene text.

Video captioning. Fig. 5 presents qualitative results from experiments on the VATEX-EVAL dataset. Cases (a) and (b) illustrate successful cases in the reference-based and reference-free settings. In Case (a), $\boldsymbol{x}_{\text{ref}}^{(1)}$ was "a young boy is on a bicycle riding in the woods and jumps a big hurdle." and x_{cand} was "a person rides a bike down a road in a mountainous area." The human judgment was 1.00 because x_{cand} appropriately describes the video. However, PAC-S++, G-VEval, and EMScore evaluated it as 0.46, 0.45 and 0.51, respectively, while the proposed metric rated it as 0.70.

Case (b) is a sample rated as moderately good by human evaluators with a score of 0.92. This score reflects a partially correct description that was semantically aligned with the video but omitted some visual details. The proposed metric rated it as 0.75, while PAC-S++ underestimated it as 0.25, G-VEval as 0.41, and EMScore as 0.32.

6 Conclusion

In this study, we addressed the task of automatic evaluation for image and video captioning and proposed RIGEL, a unified automatic evaluation metric. This metric improves LLM-based evaluation through self-distilled score adaptation framework. It resolves the mismatch between large-vocabulary language modeling and evaluation tasks that require prediction over a small set of ordinal labels. We first introduced an evaluation-specific scoring head distilled from a frozen LLM. We then refined the LLM backbone using human judgment data while freezing the head's parameters. Furthermore, we constructed Vid-Lepus, a new dataset for training human-aligned metrics for video captioning. Experiments on multiple benchmarks showed that the method achieves higher correlation with human judgments than existing methods.

7 Limitations

Although our metric achieves a high correlation with human judgments, it has the following limitations. First, Phase 1 relies on the soft pseudo-labels extracted from the frozen LM head. Improving the self-distillation procedure could further enhance the quality of the scoring head. Second, Phase 2 adapts the backbone using standard LoRA fine-tuning. However, developing adaptation methods specifically designed for evaluation tasks could yield additional gains. Third, our metric requires access to the hidden representations of the underlying LLM, which prevents the direct use of proprietary models.

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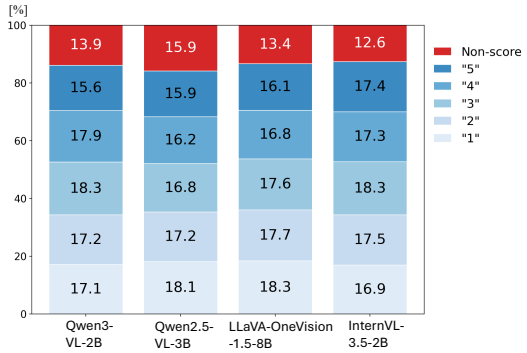


Figure 6: The logit distribution over score tokens ("1"–"5") and non-score tokens on the Composite dataset (Aditya et al., 2015). Non-score tokens exhibit logit magnitudes comparable to those of score tokens.

A Additional Related Work

Image captioning metrics. Automatic evaluation metrics for image captioning such as BLEU (Papineni et al., 2002), METEOR (Banerjee and Lavie, 2005), ROUGE (Lin, 2004), CIDEr (Vedantam et al., 2015), and SPICE (Anderson et al., 2016) have traditionally relied on reference-based lexical or semantic matching. Several extensions, such as CIDEr-R (Oliveira et al., 2021) and JaSPICE (Wada et al., 2023), have also been proposed to improve robustness or adapt evaluation to specific settings. Although these metrics remain standard in the literature, prior studies have shown that they often correlate only weakly with human judgments, especially when captions are semantically correct but lexically diverse (Hessel et al., 2021; Sarto et al., 2023, 2025).

To address this limitation, recent work has proposed data-driven metrics that leverage pre-trained vision–language models or multimodal encoders (Lee et al., 2020, 2021; Hessel et al., 2021; Sarto et al., 2023, 2025; Wada et al., 2024; Matsuda et al., 2024). Among them, CLIPScore (Hessel et al., 2021) evaluates captions by measuring image–text similarity in a reference-free manner, while PAC-S and PAC-S++ (Sarto et al., 2023, 2025) improve this paradigm by adapting CLIP-based scoring to image caption evaluation. Other approaches, such as ViLBERTScore (Lee et al., 2020), UMIC (Lee et al., 2021), Polos (Wada et al., 2024), and DENEB (Matsuda et al., 2024), further incorporate image information and/or supervised learning from human judgments. These metrics have shown strong performance on standard image captioning benchmarks, most of which primarily consist of short captions.

	Phase 1	Phase 2
Epoch	50	1 (image) / 1 (video)
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	5×10^{-4}	1×10^{-4}
Batch size	64	8
Weight decay	0.01	0.01
LoRA rank	–	8
LoRA alpha	–	16

Table 4: Settings of the proposed metric for Phases 1 and 2.

B Logit Distribution Analysis

We investigated the logit distributions over score tokens ("1"–"5") and non-score tokens as follows: For each sample in the dataset, we extracted the LM-head logits corresponding to the score tokens and aggregated the logits of all remaining vocabulary tokens into a single non-score logit using the log-sum-exp operator. We then normalized these six quantities within each sample and averaged them across samples.

Fig. 6 shows the distribution of LM-head logits assigned to score tokens ("1"–"5") and the remaining vocabulary on the Composite dataset. Non-score tokens exhibit logit magnitudes comparable to those of the score tokens across the four models (Qwen3-VL-2B (Bai et al., 2025a), Qwen2.5-VL-3B (Bai et al., 2025b), LLaVA-OneVision-1.5-8B (An et al., 2025), and InternVL-3.5-2B (Wang et al., 2025)). This observation provides partial evidence for our claim that non-score tokens function as noise in score prediction.

C Distributional Objectives for Phase 1 Distillation

We use EMD for Phase 1 distillation because the teacher distribution is defined over the ordinal five-point score space. Given two probability distributions $\boldsymbol{a}, \boldsymbol{b}$ over K ordered labels, one-dimensional EMD (Rubner et al., 2000) is defined as follows:

$$\mathrm{EMD}(\boldsymbol{a}, \boldsymbol{b}) = \sum_{i=1}^{K-1} \left| \sum_{j=1}^{i} a_j - \sum_{j=1}^{i} b_j \right|, \quad (5)$$

where a_j and b_j denote the probabilities assigned to the j-th score label. At inference, the predicted distribution is converted into a scalar score by taking the expectation over ordered labels, making label distances relevant to Phase 1 distillation. KL divergence is insufficient for this purpose because it compares the probabilities assigned to individual labels without incorporating distances in the ordinal five-point score space. In contrast, EMD penalizes discrepancies according to their distance along the ordered label axis, aligning the training objective with the expectation-based inference procedure.

Caption properties	r	ρ
Caption length	0.034	0.024
Vocabulary uniqueness	-0.017	0.010
Descriptiveness density	0.068	0.090

Table 5: Correlations between human judgments and caption properties on Vid-Lepus.

D Construction of Vid-Lepus

Statistics. The Vid-Lepus dataset consists of $3,338$ video clips extracted from the validation split of the VATEX dataset (Shi et al., 2022). Each video is paired with 10 fixed reference captions, and the number of candidate captions per video ranges from 1 to 6, with an average of 1.69. In total, the dataset contains $5,637$ candidate captions, with a vocabulary size of $7,029$ words, a total word count of $134,960$, and an average length of 23.9 words. The reference set includes $33,380$ captions, with a vocabulary size of $11,030$ words, a total word count of $804,000$, and an average length of 14.3 words. All captions are in English. The dataset also includes $14,802$ human judgments collected from 241 annotators, which is an average of 2.63 judgments per candidate caption. We confirmed that there is no overlap in videos or captions between Vid-Lepus and VATEX-EVAL, based on MD5 checksums of all video files and exact caption matching.

Annotation process. We recruited annotators from a general population on the internet via a public crowdsourcing platform, without restricting

Methods	Prompt	Composite	Flickr8K-EX	Flickr8K-CF	Nebula	VATEX-EVAL	
		τ_c	τ_c	τ_c	τ_c	τ_b	ρ
RIGEL	G-VEval	**65.6**	58.9	**20.9**	**56.4**	**42.4**	**54.0**
RIGEL	Original	64.3	**59.9**	**20.9**	55.1	42.1	53.7

Table 6: Comparison of RIGEL with the G-VEval prompt and the original prompt.

Methods	Backbone	Composite	Flickr8K-EX	Flickr8K-CF	Nebula	VATEX-EVAL	
		τ_c	τ_c	τ_c	τ_c	τ_b	ρ
G-VEval	Qwen3-VL-2B	53.1	31.5	14.8	46.2	34.0	44.1
FLEUR	Qwen3-VL-2B	60.3	46.0	18.4	51.2	–	–
RIGEL (Original)	Qwen3-VL-2B	**64.3**	**59.9**	**20.9**	**55.1**	**42.1**	**53.7**

Table 7: Comparison with existing caption evaluation methods using the same backbone.

demographic or geographic background. Annotators were recruited and compensated appropriately based on their country of residence, and consent was obtained through the task instructions, which clearly stated that the collected data would be used for research purposes.

Annotators were instructed to assess the appropriateness of candidate captions with respect to the given videos across three dimensions: descriptiveness, relevance, and fluency, each scored on a five-point scale. The scoring criteria were as follows:

5: Excellent — The caption comprehensively and accurately describes all observed objects, relationships, and contextual details, with no grammatical errors or at most one minor error.

4: Good — The caption describes most objects and relationships with only minor omissions or inaccuracies, and is generally natural and comprehensible.

3: Fair — The caption mentions key objects but lacks detail in relationships or other attributes, or contains significant inaccuracies or noticeable grammatical errors, yet remains understandable.

2: Poor — The caption includes descriptions of a few objects but omits significant details, contains numerous inaccuracies, or has errors that make it difficult to read.

1: Bad — The caption provides minimal description, is fundamentally unrelated to the video content, or contains frequent errors that render it incomprehensible.

For quality control, we excluded annotations from evaluators exhibiting suspicious behavior, such as unusually short response times or consistently uniform ratings. Furthermore, samples whose human ratings exhibited a range of at least 3 on the five-point scale are manually reviewed by an expert annotator to resolve annotation disagreements. The five-point human judgment scores were normalized to the range $[0, 1]$.

Annotation bias analysis. To investigate whether annotators systematically prioritize certain sentence structures, we conducted an additional analysis. Specifically, we computed correlations between the human judgments and three properties characterizing sentence structures on Vid-Lepus, defined in previous works (Saito et al., 2025): caption length, vocabulary uniqueness, and descriptiveness density.

Table 5 presents the correlation coefficients between the human judgments and the three properties. The table shows that caption length yielded 0.034 and 0.024 for Pearson's r and Spearman's ρ, respectively, vocabulary uniqueness yielded -0.017 and 0.010, and descriptiveness density yielded 0.068 and 0.090. These near-zero correlations indicate that annotators do not systematically prioritize particular sentence structures.

E Implementation Details

Table 4 shows the settings of the proposed metric for Phases 1 and 2. In Phase 1, the scoring head had 2.63M trainable parameters and an inference cost of 0.01 GFLOPs. In Phase 2, the model had 1.61M trainable parameters and required an average of 770 GFLOPs per inference. We trained the proposed metric on eight NVIDIA H200 SXM GPUs (141 GB VRAM per GPU) and performed evalua-

Methods	Training dataset for image captioning	Composite	Flickr8K-EX	Flickr8K-CF
		τ_c	τ_c	τ_b
Pearl	Spica	60.4	58.6	38.6
DENEB	Spica	58.6	56.9	38.2
Ours	Spica	**66.1**	**59.0**	**40.4**

Table 8: Quantitative comparison with same-data baselines on image captioning evaluation benchmarks.

tion on a single H200 GPU. The total training time was approximately 11.6 hours, and the average inference time was approximately 610 ms per sample. In Phase 1, we employed early stopping based on Kendall's τ_c. Specifically, τ_c was computed on the validation set after each epoch, and training was stopped when no improvement in τ_c was observed for five consecutive epochs. The model achieving the highest τ_c was then evaluated on the test sets. In Phase 2, the model was trained for one epoch.

F Additional Experiments

F.1 Effect of Prompt Template

We conducted controlled experiments to disentangle the effect of the prompt template. Specifically, we replaced RIGEL's original prompt template with the prompt used by G-VEval. Table 6 shows the comparison with this variant. Table 6 shows that RIGEL with the G-VEval prompt achieved performance comparable to the original RIGEL and even outperformed it on several benchmarks. This result suggests that prompt engineering could further improve the proposed metric.

F.2 Same-Backbone Comparison

Table 7 presents the quantitative comparison results with the same-backbone baselines. We implemented G-VEval and FLEUR using the same Qwen3-VL-2B backbone as RIGEL. Table 7 shows that original RIGEL outperformed the same-backbone G-VEval and FLEUR baselines across all benchmarks. These results indicate that the performance gains do not stem from the backbone or prompt engineering, but instead primarily arise from the proposed self-distillation pipeline.

F.3 Same-Data Comparison

To disentangle the contribution of the training data from that of the proposed method, we conducted controlled evaluations for both video and image captioning. Table 8 presents the quantitative results with same-data baselines in image captioning. As shown in Table 8, RIGEL outperformed both baselines on all benchmarks. Specifically, RIGEL outperformed Pearl by 5.7, 0.4, and 1.8 points on Composite, Flickr8K-Expert, and Flickr8K-CF, respectively, and outperformed DENEB trained on Spica by 7.5, 2.1, and 2.2 points on the same benchmarks, respectively.

Methods	Training dataset for video captioning	VATEX-EVAL	
		τ_b	ρ
PAC-S++	Vid-Lepus	31.2	40.3
EMScore	Vid-Lepus	30.2	39.0
Ours	Vid-Lepus	**42.1**	**53.7**

Table 9: Quantitative comparison with same-data baselines on VATEX-EVAL.

Metrics	Objective	Composite		Flickr8K-EX		Nebula		VATEX-Eval	
		τ_b	τ_c	τ_b	τ_c	τ_b	τ_c	τ_b	ρ
(i)	KL	59.0	63.8	59.3	59.7	57.7	54.7	41.9	53.4
(ii)	EMD	**59.5**	**64.3**	**59.5**	**59.9**	**58.1**	**55.1**	**42.1**	**53.7**

Table 10: **Ablation study on the Phase 1 distribution objective in the reference-free setting.** KL: Kullback–Leibler divergence; EMD: Earth Mover's Distance.

Table 9 shows the comparison with same-data baselines in video captioning. We fine-tuned PAC-S++ and EMScore on Vid-Lepus because no supervised baseline currently exists. As shown in Table 9, our proposed metric achieved 42.1 and 53.7 for τ_b and ρ, respectively, outperforming the fine-tuned PAC-S++ by 10.9 and 13.4 points and the fine-tuned EMScore by 11.9 and 14.7 points on VATEX-EVAL. These results indicate that the performance gains of RIGEL are not attributable solely to training data, but to the proposed method itself.

F.4 Objective Ablation Study

Table 10 shows the quantitative results of the ablation study. We investigated the effect of the Phase 1 distributional objective by comparing KL divergence with EMD. EMD outperformed KL divergence across all benchmarks. Specifically, EMD improved Composite by 0.5 points for both τ_b and τ_c, Flickr8K-EX by 0.2 points for both τ_b and τ_c, and Nebula by 0.4 points for both τ_b and τ_c. On VATEX-Eval, EMD improved τ_b and ρ by 0.2 and 0.3 points, respectively. These results indicate that accounting for the ordinal structure of the five-point score space provides more suitable training signals for Phase 1 distillation.

G Error Analysis

To investigate the limitations of the proposed metric, we analyzed cases where the proposed metric failed to perform as expected. We defined failure cases as samples where the absolute difference between y and $\hat{y}$ exceeded 0.5, where y denotes the normalized human judgment score. We identified 45 and 182 failure cases in the test set of the Nebula and VATEX-EVAL dataset, respectively.

Table 11 shows the results of the error analysis. We identified 6 major failure modes:

- Overestimated underspecified captions: This category includes cases where the metric gave scores that were higher than human judgment scores to candidates that were correct but lacked sufficient detail.
- Overestimated inaccurate captions: This category refers to cases where the metric gave high scores to candidates containing incorrect descriptions of objects, actions, or scenes.
- Overpenalized alternative focuses: This category includes cases where the candidate focused on visual content that differed from the references and the metric gave scores lower than the human judgments.
- Overestimated alternative focuses: This category refers to cases where the candidate focused on visual content that differed from the references and missed the main content, but the metric gave a score higher than the human judgments.
- Overpenalized local errors: This category includes cases where the metric excessively penalized candidates for minor errors, even though the overall caption remained acceptable.
- Annotation error: This category includes samples where the human judgments were inappropriate.

Table 12 shows the results of the error analysis on VATEX-EVAL. Among the 182 VATEX-EVAL failure cases, we manually analyzed the 100 cases with the largest absolute errors. We identified 6 major failure modes:

- Underestimated accurate captions: This category encompasses cases where the proposed metric gave scores lower than those of human judgment to candidates that accurately describe the video.

Description	Count
Overestimated underspecified captions	14
Overestimated inaccurate captions	8
Overpenalized alternative focuses	2
Overestimated alternative focuses	6
Overpenalized local errors	2
Annotation error	13
Total	45

Table 11: Categorization of the failure modes on Nebula. We analyzed the 45 samples with the greatest absolute differences between $\hat{y}$ and y.

Description	Count
Underestimated accurate captions	29
Underestimated lexical variants	43
Underestimated alternative focuses	16
Overestimated inaccurate captions	3
Overpenalized local errors	4
Annotation error	5
Total	100

Table 12: Categorization of the failure modes on VATEX-EVAL. We analyzed the 100 samples with the greatest absolute differences between $\hat{y}$ and y.

- Underestimated lexical variants: This category refers to cases where the proposed metric underestimated candidates that used expressions different from the references, even though they preserved the meaning.
- Underestimated alternative focuses: This category includes cases where the proposed metric gave lower scores to candidates that focused on valid scenes, objects, or actions different from those emphasized in the references.
- Overestimated inaccurate captions: This category refers to cases where the proposed metric gave scores higher than the human judgments to captions containing incorrect descriptions.
- Overpenalized local errors: This category includes cases where the proposed metric excessively penalized captions for minor errors, even when the overall caption remained largely appropriate.
- Annotation error: This category includes samples where the human judgments were inappropriate.

Table 12 shows that the primary cause of errors was underestimation, mainly caused by the limited coverage of the reference captions. These errors likely arise because the metric places strong emphasis on references and does not fully exploit visual evidence in this setting. Consequently, visually consistent candidates can be underestimated when they differ from the references in wording or focus. In future work, we plan to extend the metric by introducing a visual-token compression mechanism that enables the model to incorporate richer visual evidence (Chai et al., 2025; Zeng et al., 2025).

H Prompts in RIGEL

This section provides the full prompts for all four settings: reference-free and reference-based video captioning, and reference-free and reference-based image captioning.

Full prompt for video captioning in the reference-free setting

Evaluate the quality of a video caption based on video frames.

Evaluation Criteria:
- Score ranges from 1 to 5
- 1: Completely incorrect or irrelevant
- 5: Perfectly describes the video content
- Penalize redundancies and irrelevant information
- Score 1 if the generated caption contains any hallucination

Evaluation Steps:
1. Examine the video frames to understand the main content
2. Assess how accurately the caption describes the video
3. Compare the generated caption to video frames
4. Assess coverage of main points and relevance
5. Assign ONE score from 1 to 5

Generated Caption:
cand

Please output only a single integer from 1 to 5, without any explanation or formatting.
Your score is

Full prompt for video captioning in the reference-based setting

Evaluate the quality of a video caption based on video frames and reference captions.

Evaluation Criteria:
- Score ranges from 1 to 5
- 1: Completely incorrect or irrelevant
- 5: Perfectly describes the video content with all key information
- Penalize redundancies and irrelevant information
- Score 1 if the generated caption contains any hallucination

Evaluation Steps:
1. Examine the video frames to understand the main content
2. Assess how accurately the caption describes the video
3. Compare the generated caption to both video frames and references
4. Assess coverage of main points and relevance
5. Assign ONE score from 1 to 5

Reference Captions:
refs_text
Generated Caption:
cand
Please output only a single integer from 1 to 5, without any explanation or formatting.
Your score is

Full prompt for image captioning in the reference-free setting

Evaluate the quality of a image caption based on image.

Evaluation Criteria:
- Score ranges from 1 to 5
- 1: Completely incorrect or irrelevant
- 5: Perfectly describes the image content
- Penalize redundancies and irrelevant information
- Score 1 if the generated caption contains any hallucination

Evaluation Steps:
1. Examine the image to understand the main content

2. Assess how accurately the caption describes the image
3. Compare the generated caption to image
4. Assess coverage of main points and relevance
5. Assign ONE score from 1 to 5
Generated Caption:
cand
Please output only a single integer from 1 to 5, without any explanation or formatting.
Your score is

Full prompt for image captioning in the reference-based setting

Evaluate the quality of a image caption based on image and reference captions.

Evaluation Criteria:
- Score ranges from 1 to 5
- 1: Completely incorrect or irrelevant
- 5: Perfectly describes the image content with all key information
- Penalize redundancies and irrelevant information
- Score 1 if the generated caption contains any hallucination

Evaluation Steps:
1. Examine the image to understand the main content
2. Assess how accurately the caption describes the image
3. Compare the generated caption to both image and references
4. Assess coverage of main points and relevance
5. Assign ONE score from 1 to 5

Reference Captions:
refs_text

Generated Caption:
cand

Please output only a single integer from 1 to 5, without any explanation or formatting.
Your score is

I Additional Details for ARR Checklist

Discuss the License for Artifacts. RIGEL and Vid-Lepus are released under the BSD 3-Clause Clear License. The licenses of the models and datasets used in this study are summarized below:

Qwen3-VL (Bai et al., 2025a):
Apache 2.0 license

Qwen2.5-VL (Bai et al., 2025b):
Qwen Research License Agreement

LLaVA-OneVision-1.5 (An et al., 2025):
Apache 2.0 license

InternVL-3.5 (Wang et al., 2025):
Apache 2.0 license

Spica (Hirano et al., 2026):
License not explicitly specified by the distributor

Flickr8K-Expert (Hodosh et al., 2013):
License not explicitly specified by the distributor

Flickr8K-CF (Hodosh et al., 2013):
License not explicitly specified by the distributor

Nebula (Matsuda et al., 2024):
License not explicitly specified by the distributor

Composite (Aditya et al., 2015):
License not explicitly specified by the distributor

FOIL (Shekhar et al., 2017):
Creative Commons Attribution 4.0 license

VATEX-EVAL (Shi et al., 2022):
MIT license

ActivityNet-Fact (Liu and Wan, 2023):
License not explicitly specified by the distributor

YouCook2-Fact (Liu and Wan, 2023):
License not explicitly specified by the distributor

ActivityNet-FOIL (Shi et al., 2022):
MIT license

Artifact Use Consistent With Intended Use. All existing artifacts used in this study were employed in accordance with their intended use. For the artifacts developed in this study, we define their intended use as general academic and research purposes, consistent with the original access conditions of the datasets and models used.

Data Contains Personally Identifying Info Or Offensive Content. The collected data do not contain personally identifiable or offensive content. All data used in this study are publicly available. We further confirmed that the source websites, repositories, and publications contain no statements raising concerns about personal information.